

MAT 310, Handout 3: Combinatorics (2.6)

mn Rule: Given set A (with m elements) and set B (with n elements), the number of unique pairs that can be selected with one element from A and one element from B is _____.

Note: the mn Rule generalizes to 3 or more dimensions.

Example 1. Find the number of combinations of:

- (a) a pair of initials
- (b) three initials
- (c) standard issue KY license plates (3 initials, 3 numeric digits)

Example 2. How many ways are there to arrange a batting lineup of 9 baseball players?

Permutation: P_r^n is the # of ways to fill r unique slots out of n (for $n \geq r$) objects.

$$P_r^n =$$

Note: $0! =$

Example 3. If 4 bedrooms are in an apartment and each tenant gets their own bedroom, find the # of ways of arranging tenants into bedrooms if there are:

- (a) 4 tenants
- (b) 9 potential tenants

Question: What if the objects don't need to be ordered, but just selected?

Combination: $C_r^n = \binom{n}{r}$ is the # of ways of selecting r objects out of a group of n distinct objects, i.e., it is the number of subsets of size r out of the n objects.

$$C_r^n = \binom{n}{r} =$$

Example 4. 16 students try out for a 12 player basketball team. How many different roster combinations are there?

How does this compare to the number of ways to select the 4 players not on the roster?

We generalize: $C_r^n = C^n_r$, that is, $\binom{n}{r} = \binom{n}{n-r}$.

The number of ways to partition n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects each, such that $\sum_{i=1}^k n_i = n$, is given by:

$$\binom{n}{n_1, n_2, \dots, n_k} =$$

Example 5. Suppose there are 10 safety officers that need to be divided among three shifts, so that 4 are working in the 1st shift, 3 in the 2nd shift, and 3 in the 3rd shift. How many possible ways are there to make this division?

Example 6. There are 8 seats in the first row at a ball game, labeled 1 through 8. One person will be assigned to each seat.

- (a) How many possible arrangements are there?
- (b) How many if Jack and Jill must sit together?
- (c) How many if there are 4 couples who plan to sit with each other?
- (d) How many if there are 4 guys who plan to sit together, and 4 girls who plan to sit together?

Example 7. 40 players are on a football team. We want to divide the players into an offensive unit (18 players), defensive unit (19 players), and special teams (3 players).

- (a) How many different divisions are possible?
- (b) What is the probability that two brothers (2 of the 40 players) are randomly selected to play on the same unit?

Example 8. Suppose you have 2 blue marbles, 2 green marbles, and 2 red marbles in a bag.

- (a) If you randomly select 2 marbles at random, what is the probability that you select different colors?
- (b) If you select 3 marbles at random, what is the probability that you select exactly 2 colors?

Example 9. Given a typical 52 card deck, find the probability of being dealt the following 5-card hands:

- (a) flush (all same suit)
- (b) one pair (no better)
- (c) two pairs (no better)
- (d) 3 of a kind (no better)